Thornquist Outfitters Data Lake

Evidence of Implementation (Screenshots)

CCGC 5506 Midterm, Project 1

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AWS Learner Lab, us-east-1 · Bucket: thornquist-lake-764541939057

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1. Architecture

Overall design of the governed data lake.

Architecture Diagram

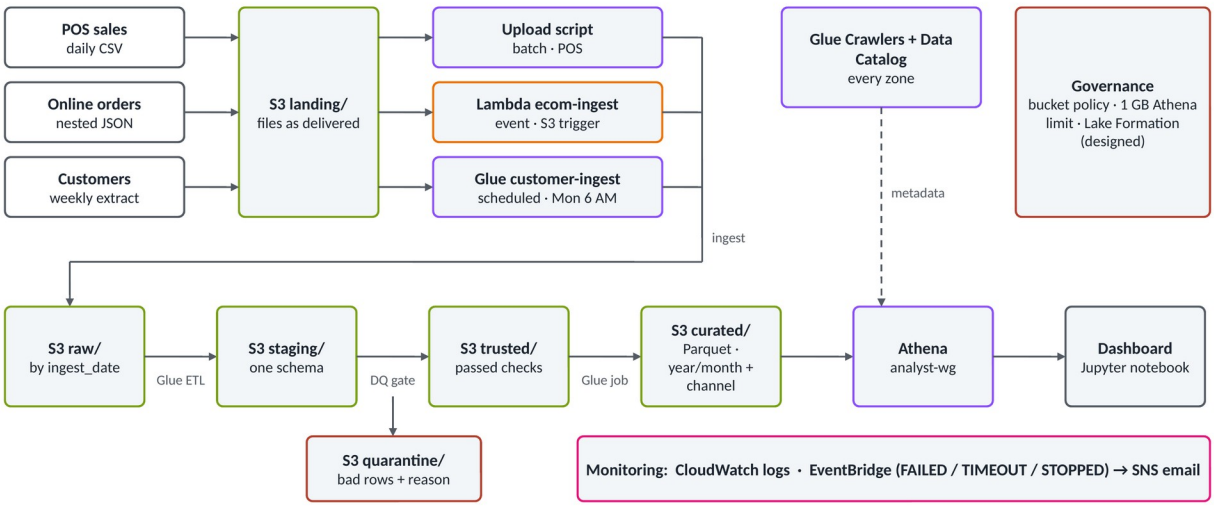


Figure 1. Architecture diagram: sources, three ingestion patterns, the S3 zones, catalog, Athena, monitoring and governance

2. Environment and S3 zoning

AWS Learner Lab (us-east-1, account 764541939057); one bucket with a prefix per zone.

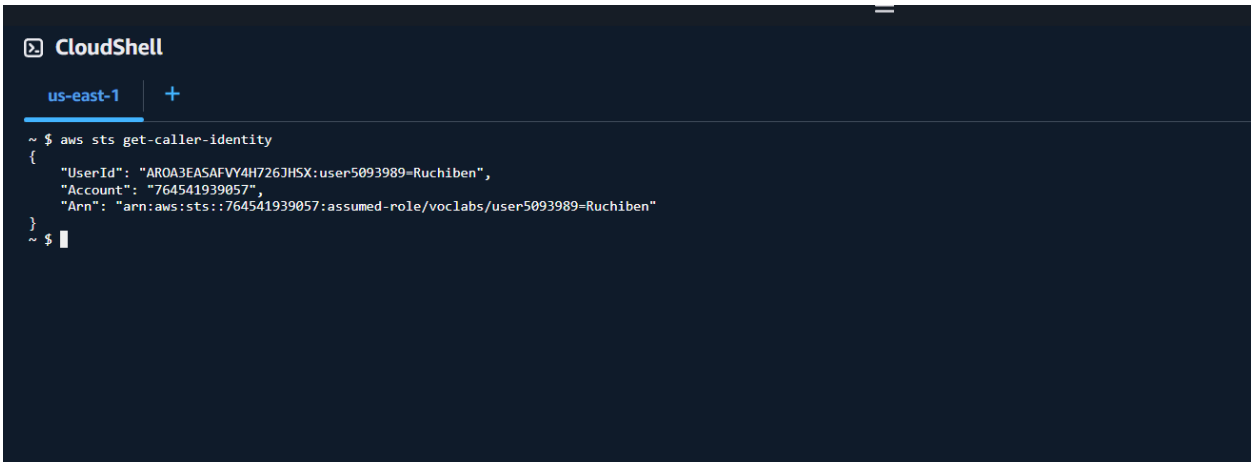


Figure 2. CloudShell identity check: account 764541939057, role voclabs/user5093989=Ruchiben



Figure 3. Zone prefixes created in thornquist-lake-764541939057

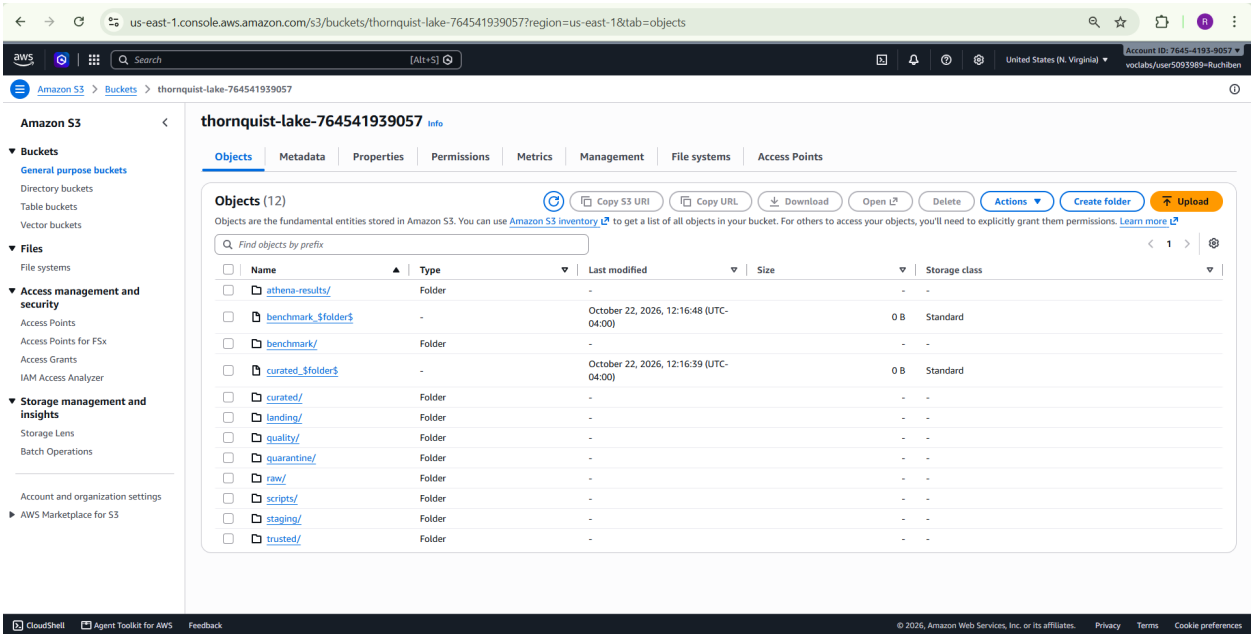


Figure 4. S3 console: all zone folders, including quality/, curated/ and benchmark/

3. Ingestion (three patterns)

Event-driven (Lambda), batch upload script (POS), and a scheduled Glue job (customers).

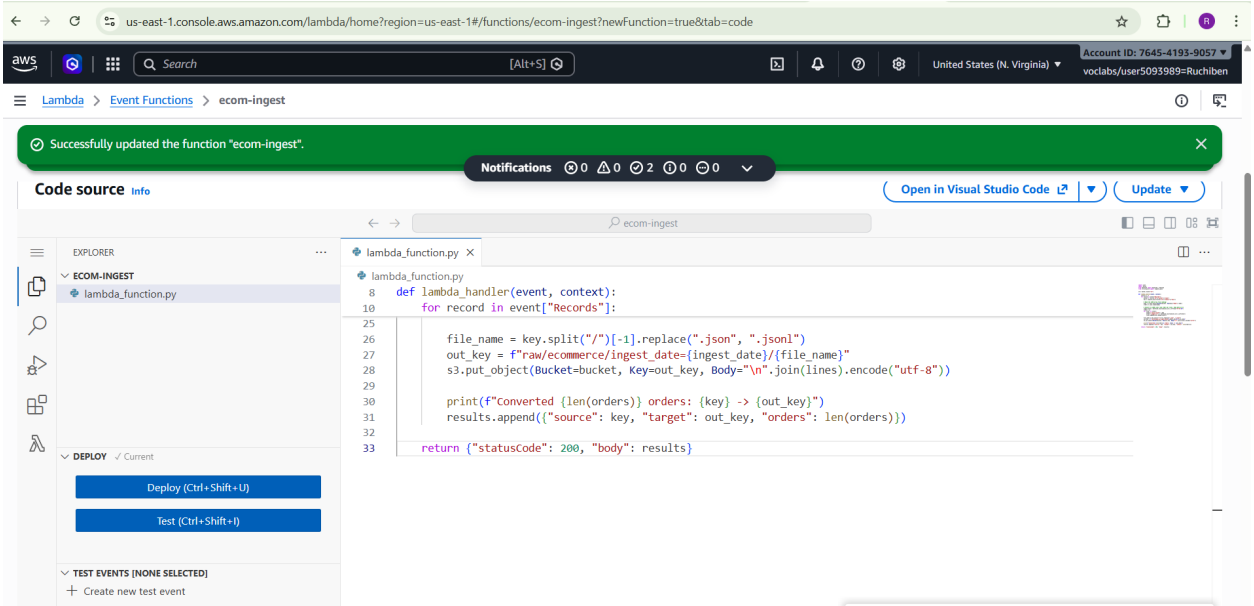


Figure 5. Lambda ecom-ingest code: converts a JSON array into JSON Lines and adds _source_file and _ingest_ts

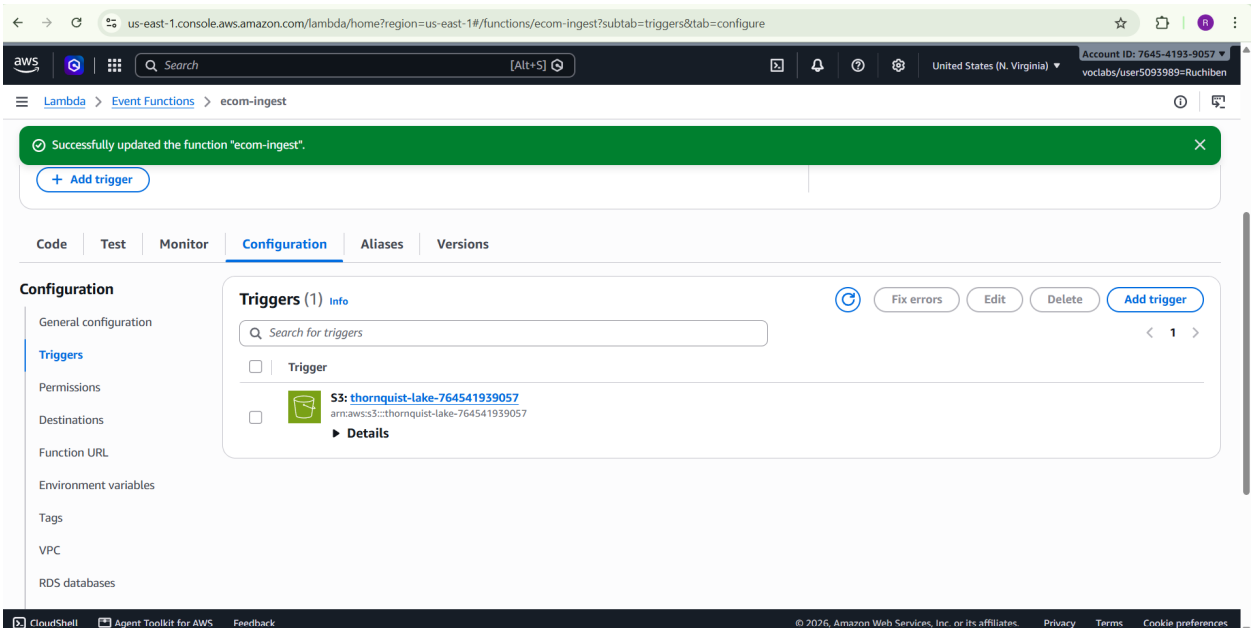


Figure 6. S3 trigger on landing/ecommerce/ with .json suffix (prevents the Lambda triggering itself)



Figure 7. Lambda output written to raw/ecommerce/ingest_date=.../



Figure 8. POS file loaded from landing into raw/pos by the upload script

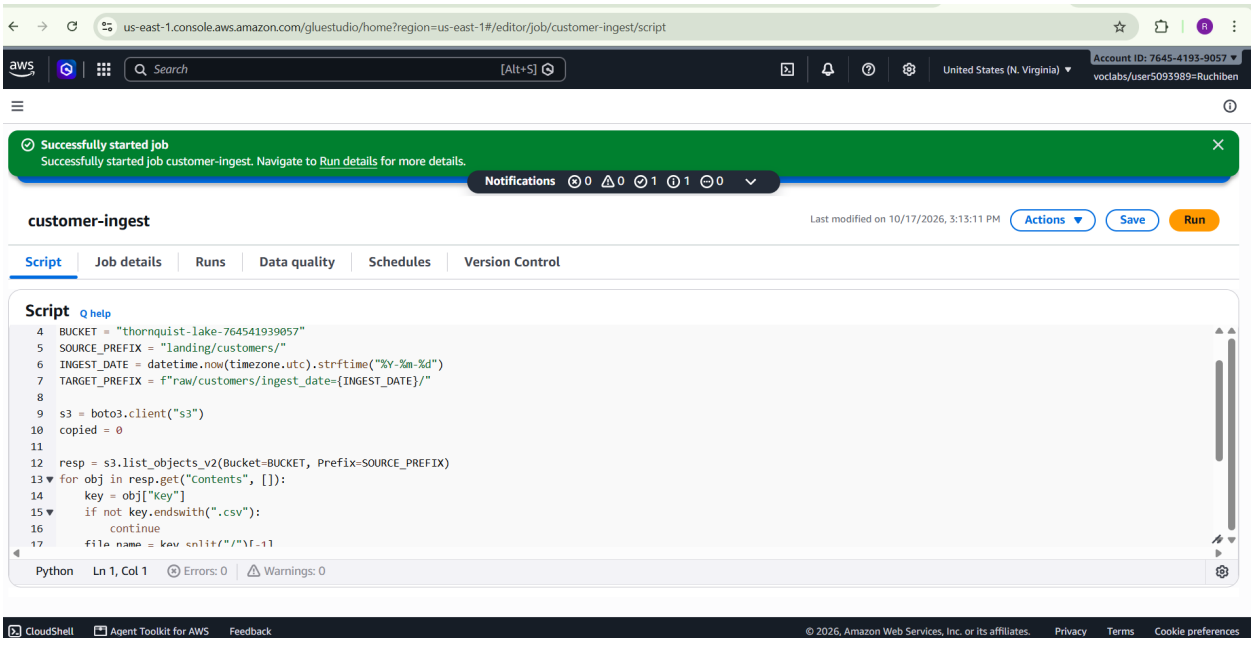


Figure 9. Glue Python-shell job customer-ingest



Figure 10. Customer extract written to raw/customers/ingest_date=.../

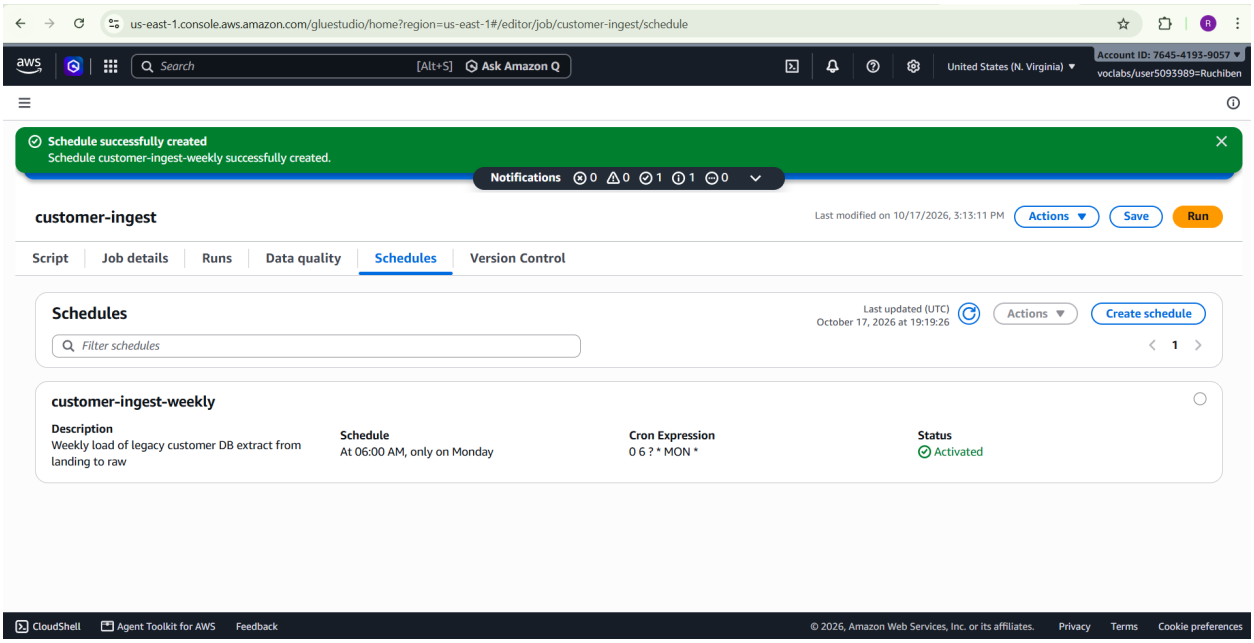


Figure 11. Schedule customer-ingest-weekly: every Monday 06:00 UTC

4. Data Catalog and raw counts

Crawlers register the raw zone; Athena confirms nothing was lost.

```
~ $ export BUCKET=thornquist-lake-764541939057
~ $ aws s3 ls s3://$BUCKET/raw/ --recursive
2026-10-17 17:05:45          0 raw/customers/
2026-10-17 19:13:30      32519 raw/customers/ingest_date=2026-10-17/customer_extract.csv
2026-10-17 17:05:45          0 raw/ecommerce/
2026-10-17 19:06:08     131293 raw/ecommerce/ingest_date=2026-10-17/ecommerce_orders.jsonl
2026-10-17 17:05:44          0 raw/pos/
2026-10-17 19:08:01     103415 raw/pos/ingest_date=2026-10-17/pos_sales_2026-08-24_to_2026-08-30.csv
~ $ aws glue update-crawler --name raw-crawler \
>   --role arn:aws:iam::764541939057:role/LabRole \
>   --targets '{"S3Targets":[{"Path":"s3://thornquist-lake-764541939057/raw/ecommerce/"},{"Path":"s3://thornquist-lake-764541939057/raw/customers/"}]}'
~ $ aws glue start-crawler --name raw-crawler
~ $ aws glue get-crawler --name raw-crawler --query 'Crawler.{State:State,LastRun:LastCrawl.Status}'
{
    "State": "RUNNING",
    "LastRun": "SUCCEEDED"
}
~ $ aws glue get-tables --database-name thornquist_raw --query 'TableList[].Name'
[
    "customers",
    "ecommerce"
]
~ $
```

Figure 12. raw-crawler succeeded; tables registered in thornquist_raw

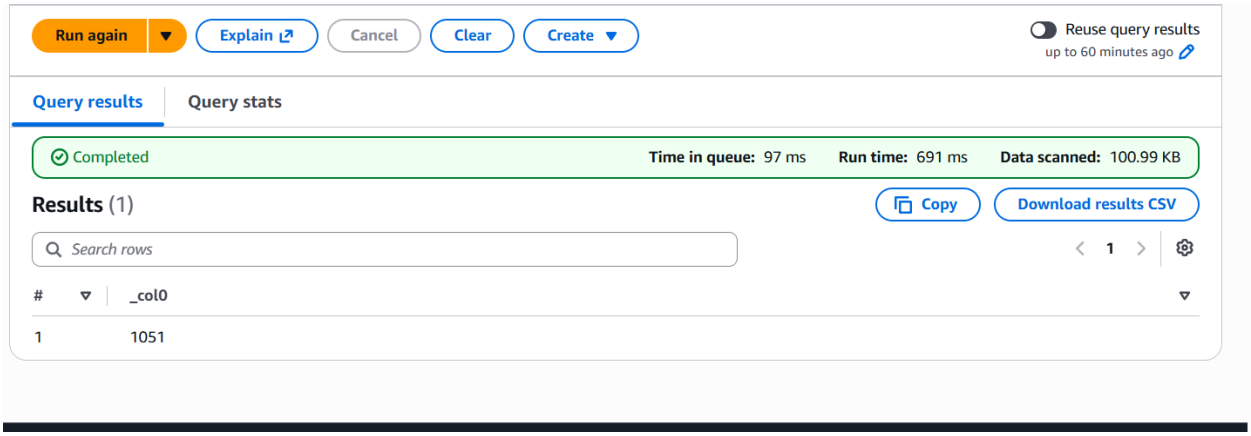


Figure 13. Raw POS row count: 1,051

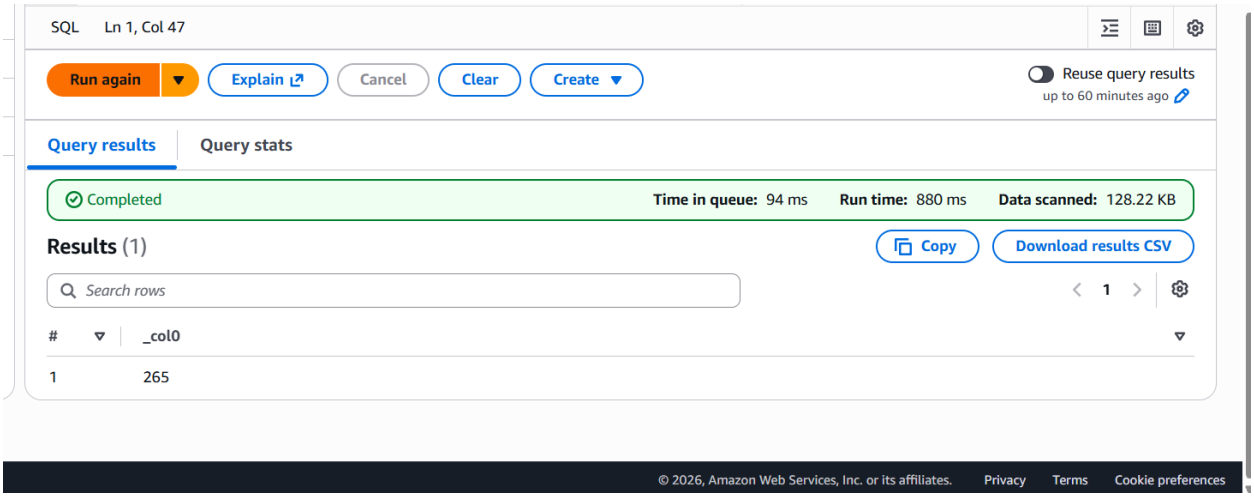


Figure 14. Raw e-commerce order count: 265

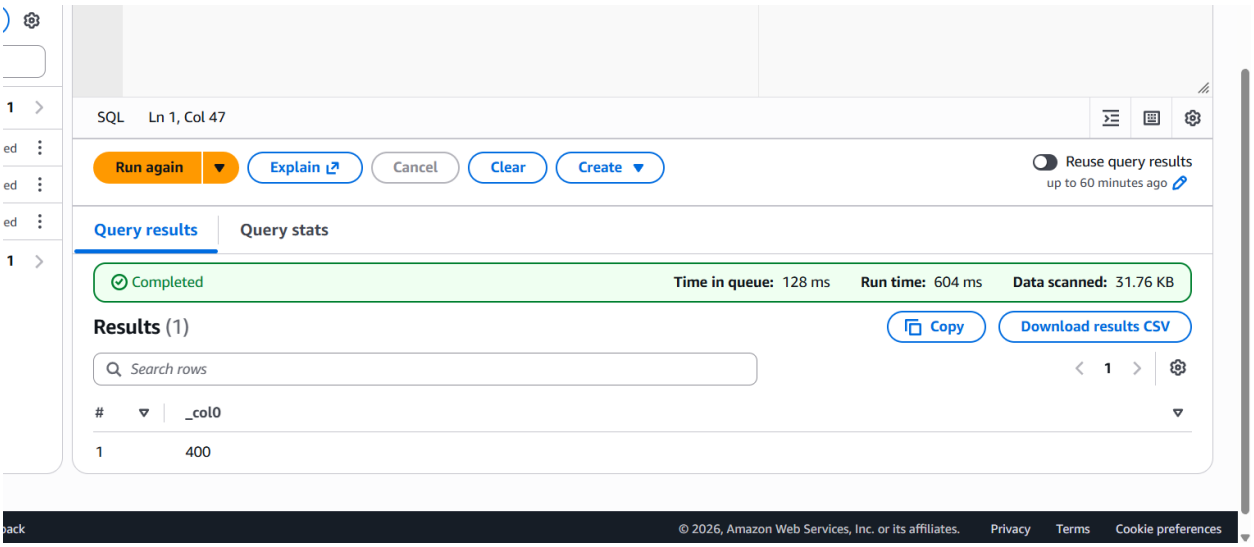


Figure 15. Raw customer count: 400

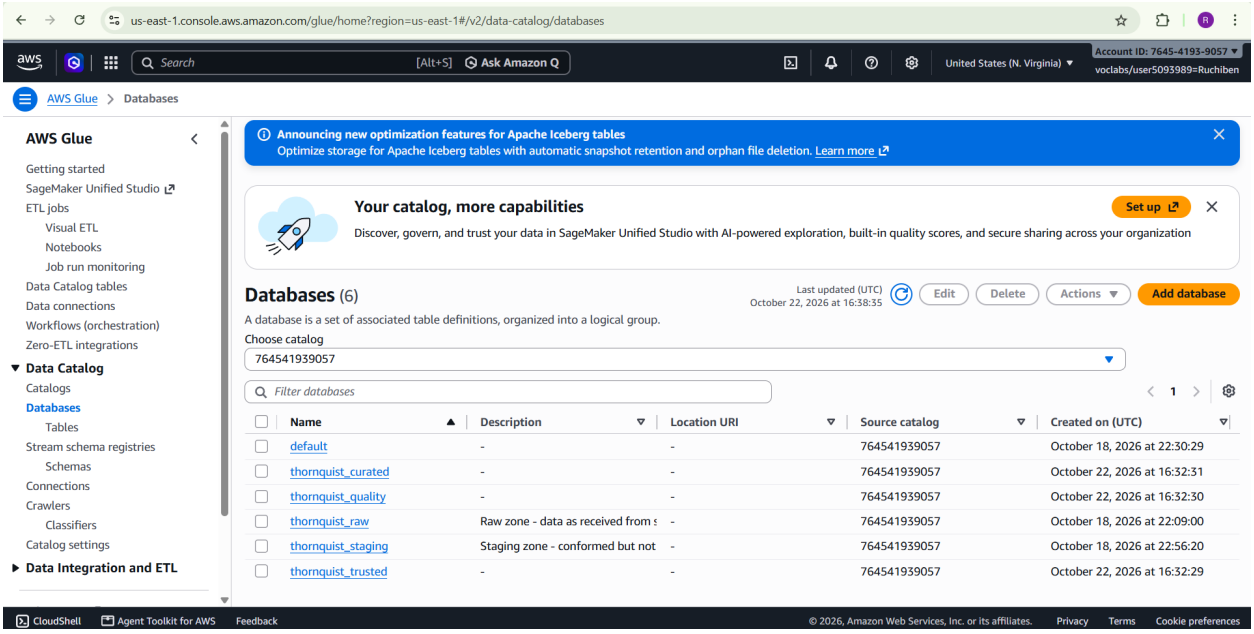


Figure 16. Glue Data Catalog databases covering every zone (raw, staging, trusted, quality, curated)

5. ETL: raw to staging

Glue Spark job raw-to-staging fixes 6 schema conflicts and writes one conformed table.



Figure 17. Staging output files (Parquet) written by raw-to-staging

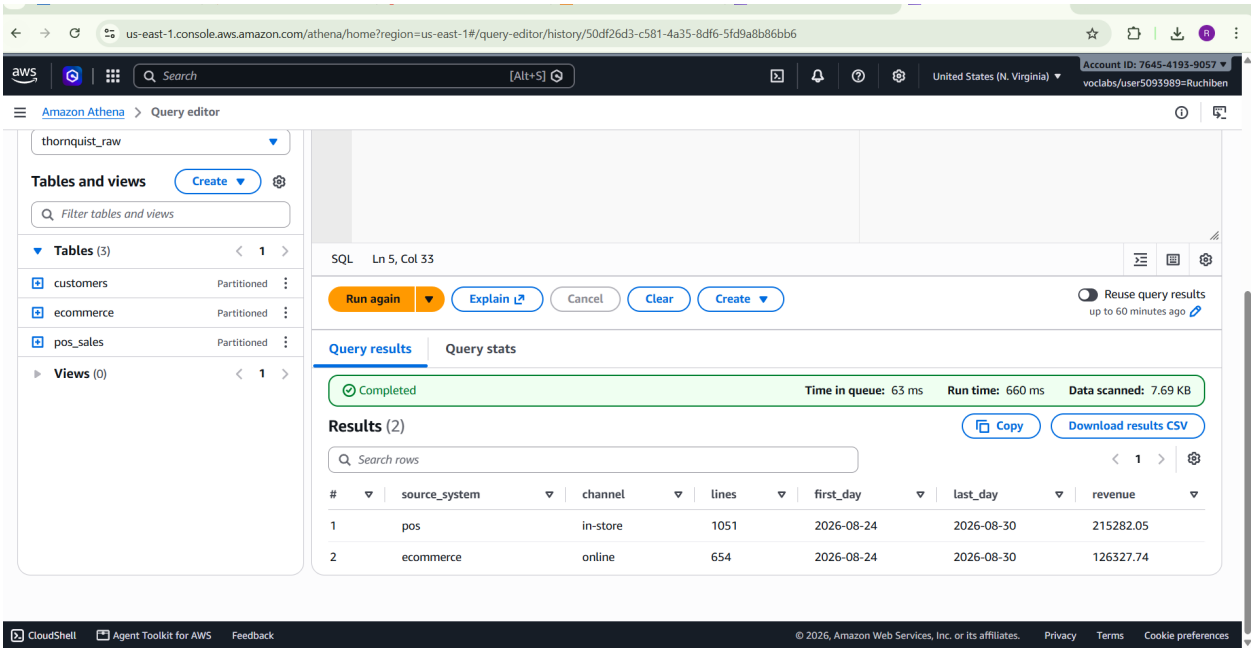


Figure 18. Staging by channel: in-store 1,051 lines / $215,282.05, online 654 lines / $126,327.74

6. Monitoring and alerting

EventBridge rule glue-job-failure (FAILED / TIMEOUT / STOPPED) sends to SNS topic thornquist-pipeline-alerts.



Figure 19. Alert test: customer-ingest started and stopped to trigger the rule



Figure 20. Live alert test repeated for the demo recording

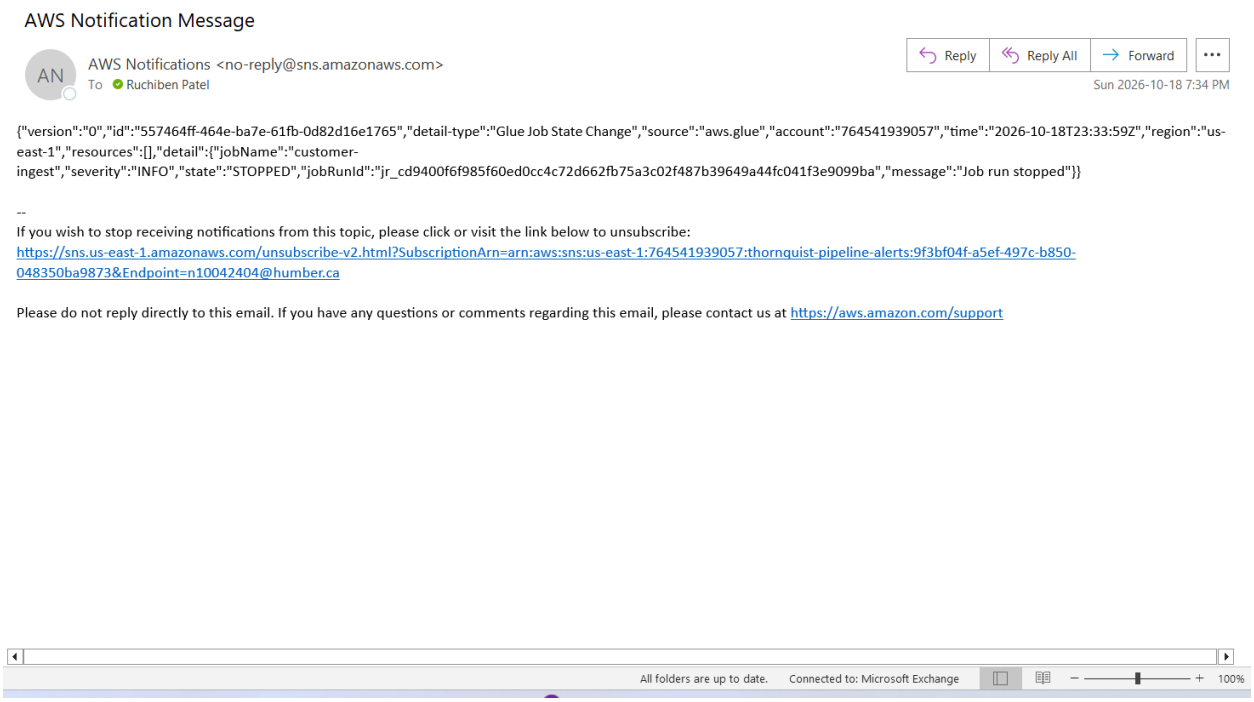


Figure 21. SNS alert email received: Glue Job State Change, customer-ingest, state STOPPED

7. Data quality gate

Glue job dq-gate: 1,705 lines checked, 1,601 passed to trusted, 104 quarantined with a reason.

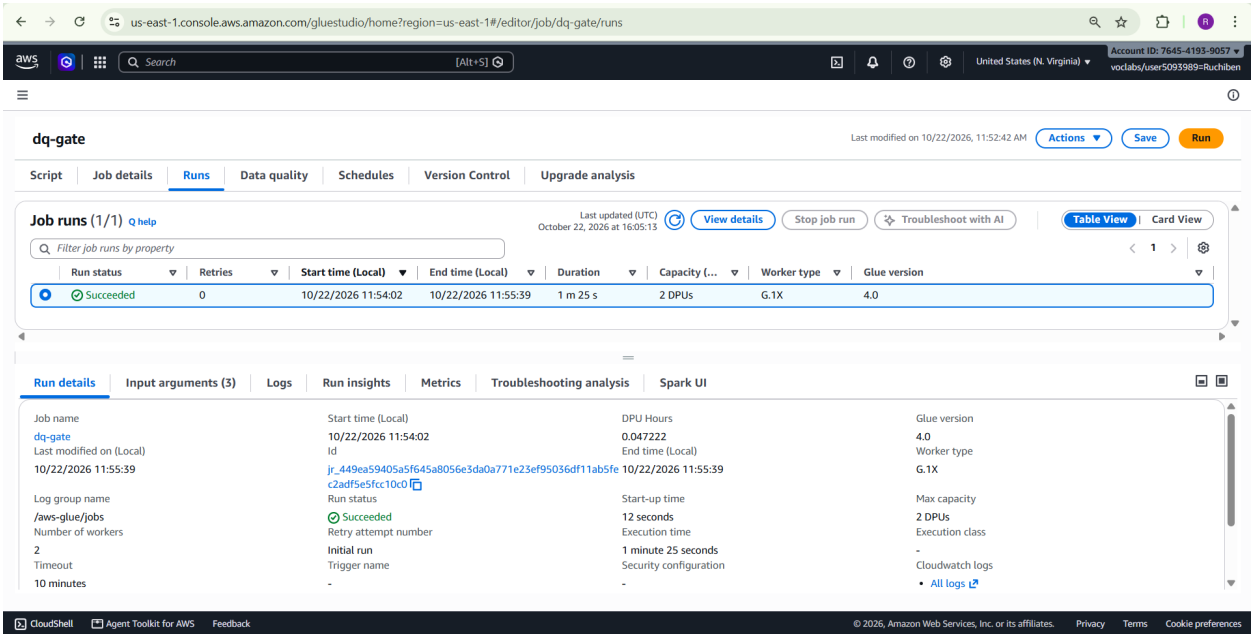


Figure 22. dq-gate job run: Succeeded

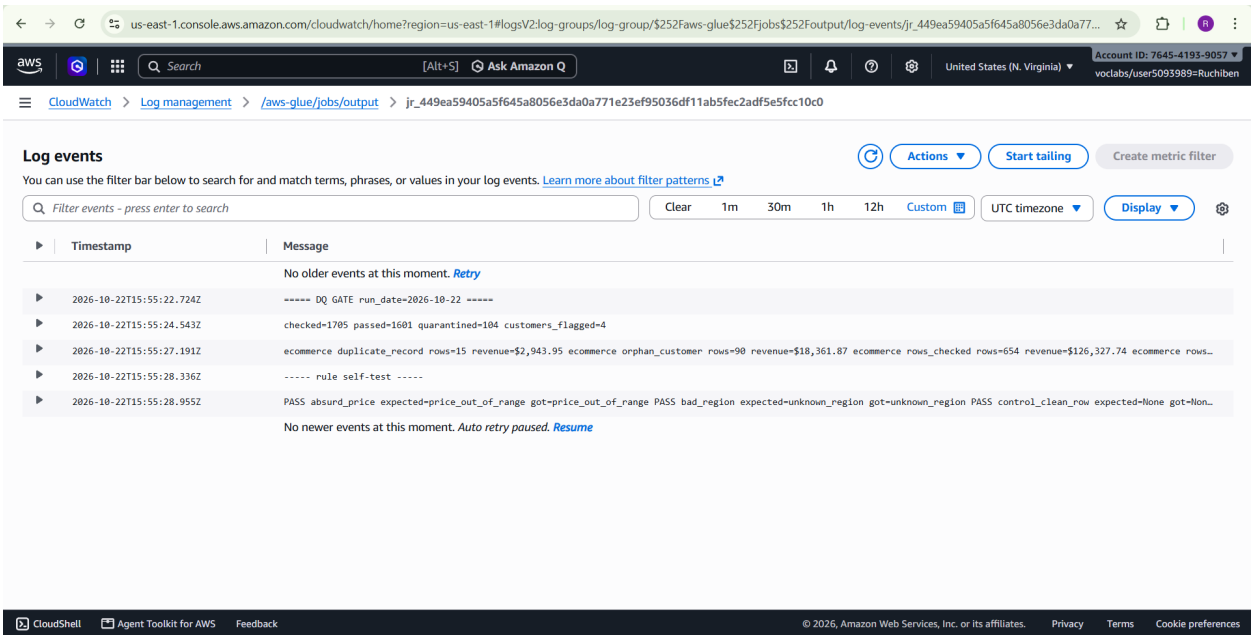


Figure 23. dq-gate log: rows checked, passed and quarantined

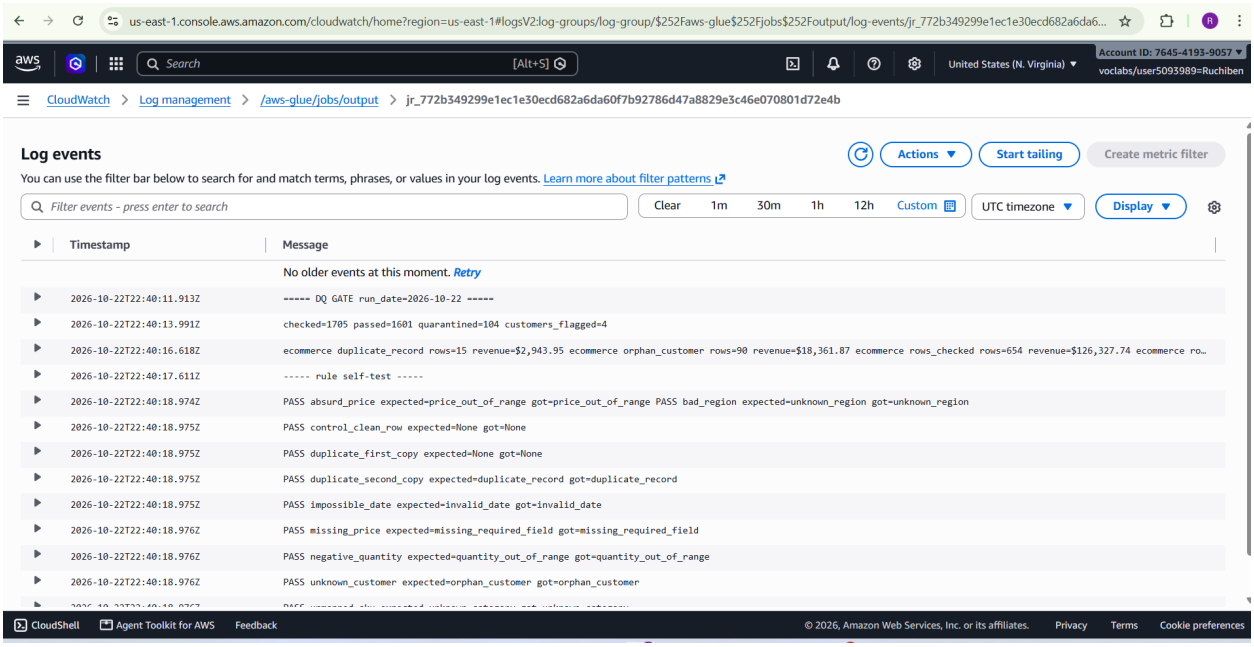


Figure 23b. dq-gate log after the demo cleanup: same results (1,601 passed, 104 quarantined, 4 customers flagged), all 10 rule self-tests PASS

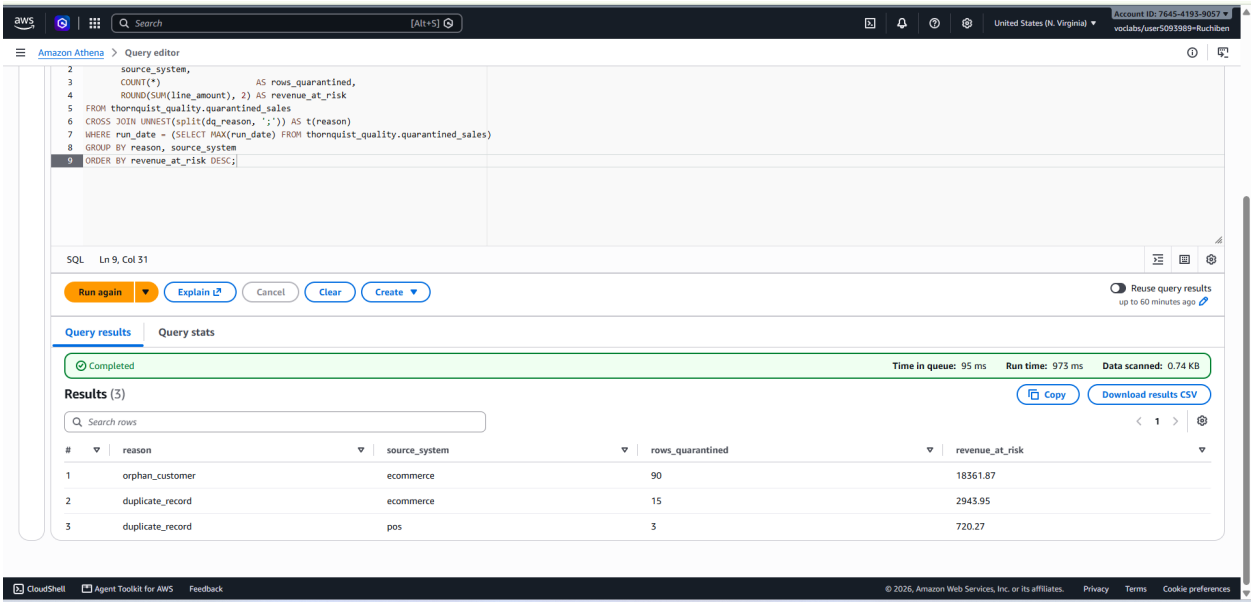


Figure 24. QA1, quarantine by reason: orphan customers 90 rows ($18,361.87), duplicates 15 online + 3 POS

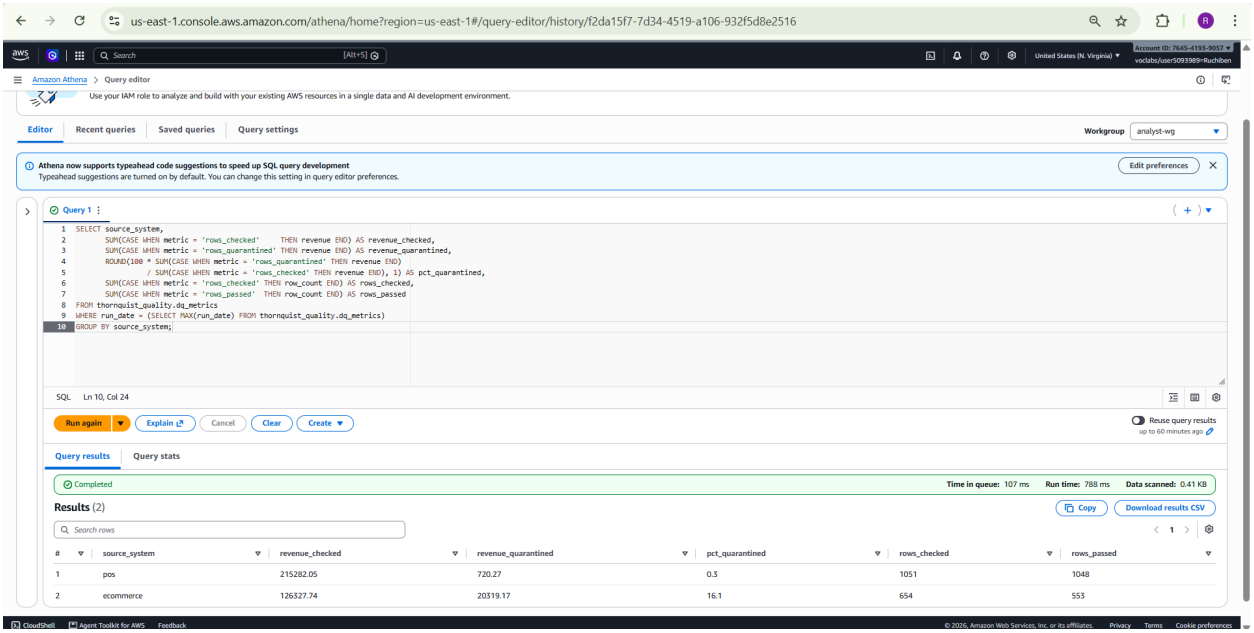


Figure 25. QA2, cost of bad data: 0.3% of in-store vs 16.1% of online revenue quarantined

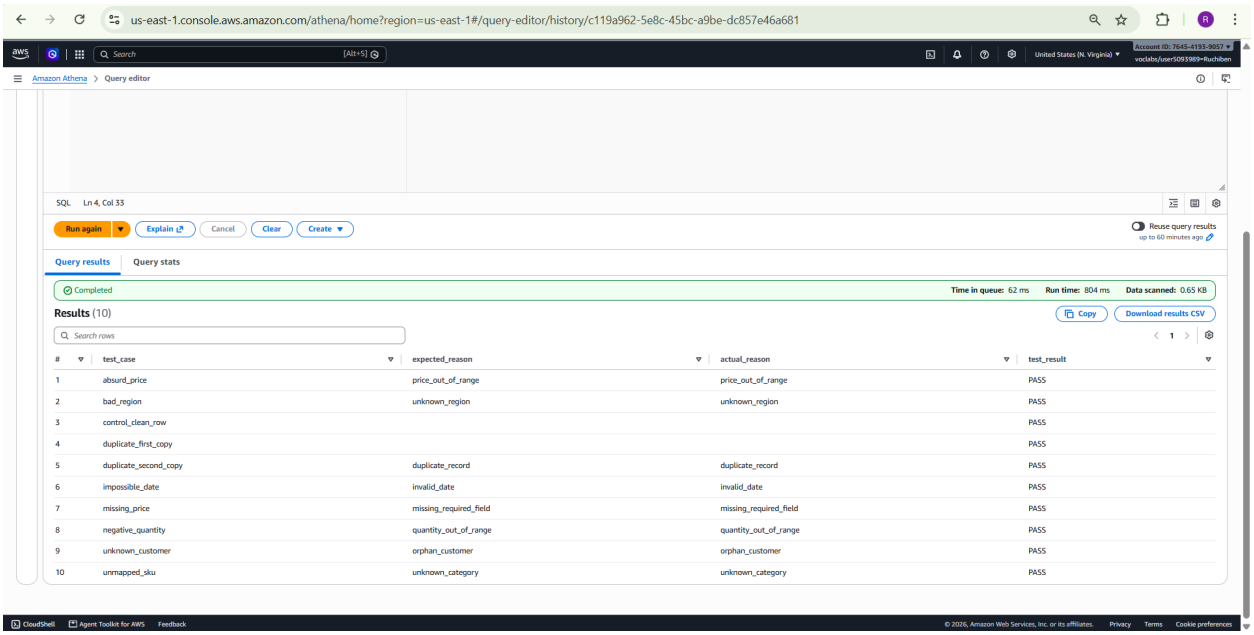
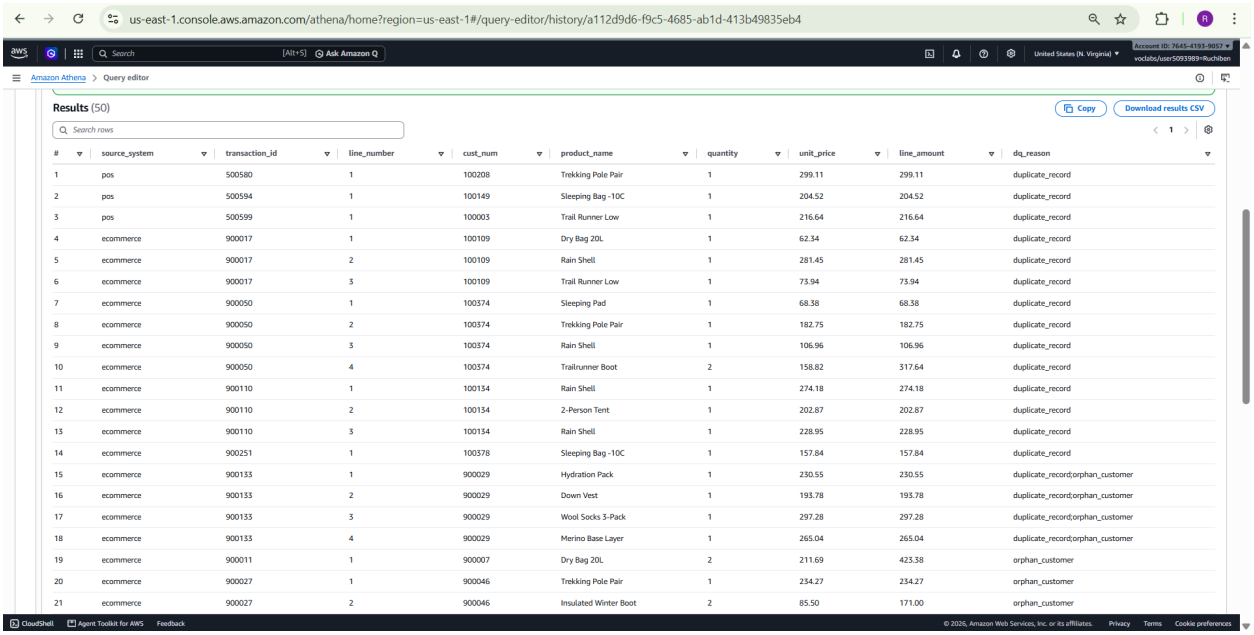


Figure 26. QA3, rule self-test: 10/10 PASS



#	source_system	transaction_id	line_number	cust_num	product_name	quantity	unit_price	line_amount	dq_reason
1	pos	500580	1	100208	Trekking Pole Pair	1	299.11	299.11	duplicate_record
2	pos	500594	1	100149	Sleeping Bag -10C	1	204.52	204.52	duplicate_record
3	pos	500599	1	100003	Trail Runner Low	1	216.64	216.64	duplicate_record
4	ecommerce	900017	1	100109	Dry Bag 20L	1	62.34	62.34	duplicate_record
5	ecommerce	900017	2	100109	Rain Shell	1	281.45	281.45	duplicate_record
6	ecommerce	900017	3	100109	Trail Runner Low	1	73.94	73.94	duplicate_record
7	ecommerce	900050	1	100374	Sleeping Pad	1	68.38	68.38	duplicate_record
8	ecommerce	900050	2	100374	Trekking Pole Pair	1	182.75	182.75	duplicate_record
9	ecommerce	900050	3	100374	Rain Shell	1	106.96	106.96	duplicate_record
10	ecommerce	900050	4	100374	Trailrunner Boot	2	158.82	317.64	duplicate_record
11	ecommerce	900110	1	100134	Rain Shell	1	274.18	274.18	duplicate_record
12	ecommerce	900110	2	100134	2-Person Tent	1	202.87	202.87	duplicate_record
13	ecommerce	900110	3	100134	Rain Shell	1	228.95	228.95	duplicate_record
14	ecommerce	900251	1	100378	Sleeping Bag -10C	1	157.84	157.84	duplicate_record
15	ecommerce	900133	1	900029	Hydration Pack	1	230.55	230.55	duplicate_record;orphan_customer
16	ecommerce	900133	2	900029	Down Vest	1	193.78	193.78	duplicate_record;orphan_customer
17	ecommerce	900133	3	900029	Wool Socks 3-Pack	1	297.28	297.28	duplicate_record;orphan_customer
18	ecommerce	900133	4	900029	Merino Base Layer	1	265.04	265.04	duplicate_record;orphan_customer
19	ecommerce	900011	1	900007	Dry Bag 20L	2	211.69	423.38	orphan_customer
20	ecommerce	900027	1	900046	Trekking Pole Pair	1	234.27	234.27	orphan_customer
21	ecommerce	900027	2	900046	Insulated Winter Boot	2	85.50	171.00	orphan_customer

Figure 27. QA4, rejected rows with their dq_reason

8. Curated layer

Glue job trusted-to-curated writes Parquet partitioned by year/month and channel.

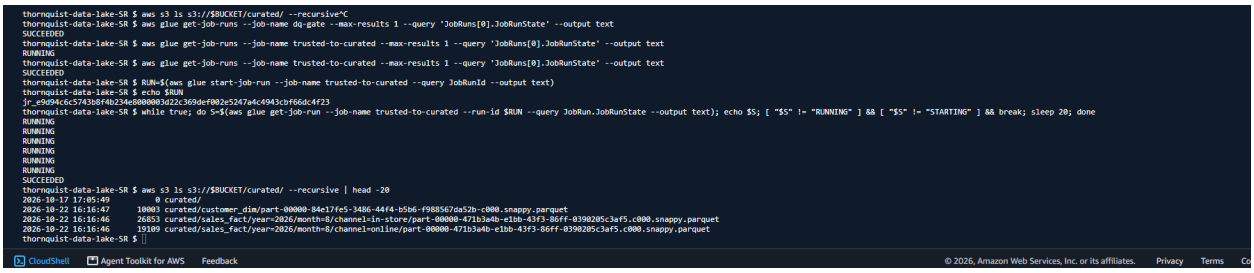


Figure 28. Curated sales_fact partitions: year=2026/month=8/channel=in-store and channel=online

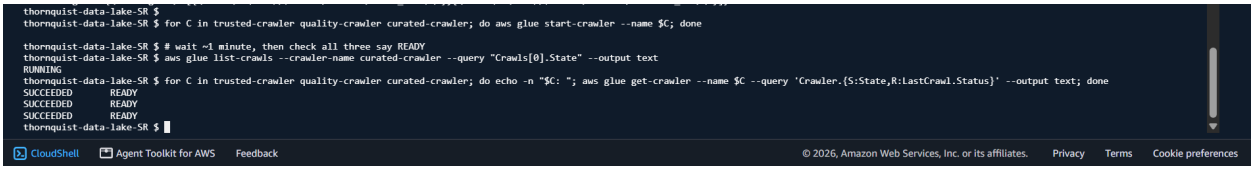


Figure 29. trusted, quality and curated crawlers: READY

9. Business analytics (Athena, workgroup analyst-wg)

Six business questions answered from the curated tables.

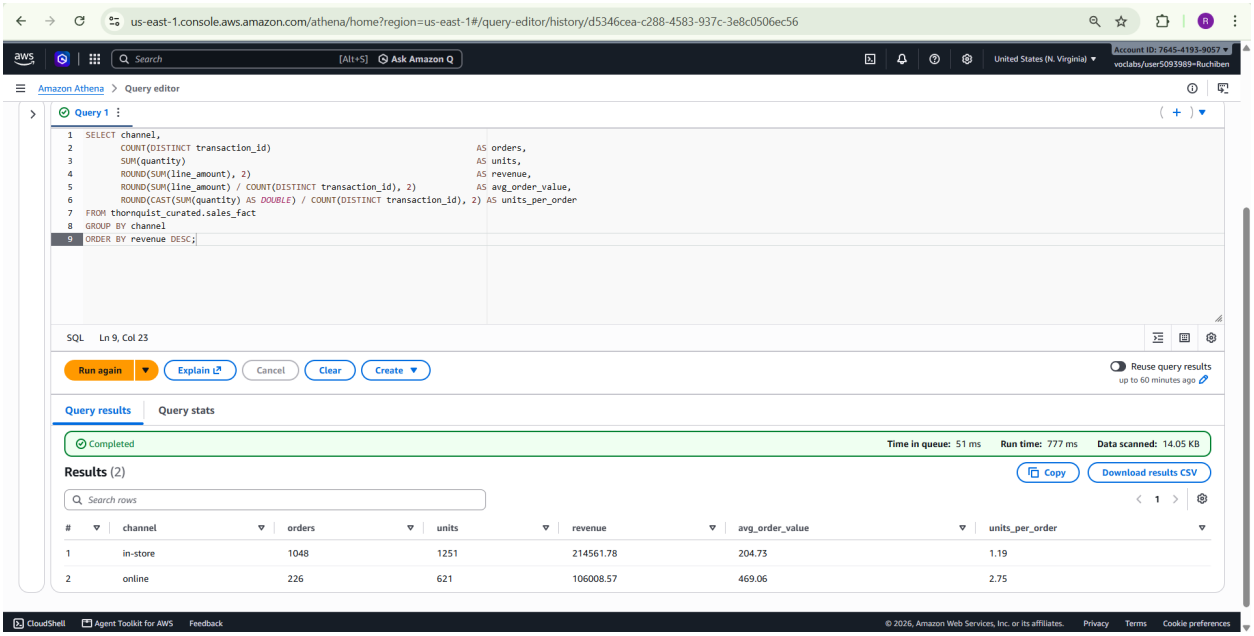


Figure 30. Q1, order value by channel: online $469.06 vs in-store $204.73

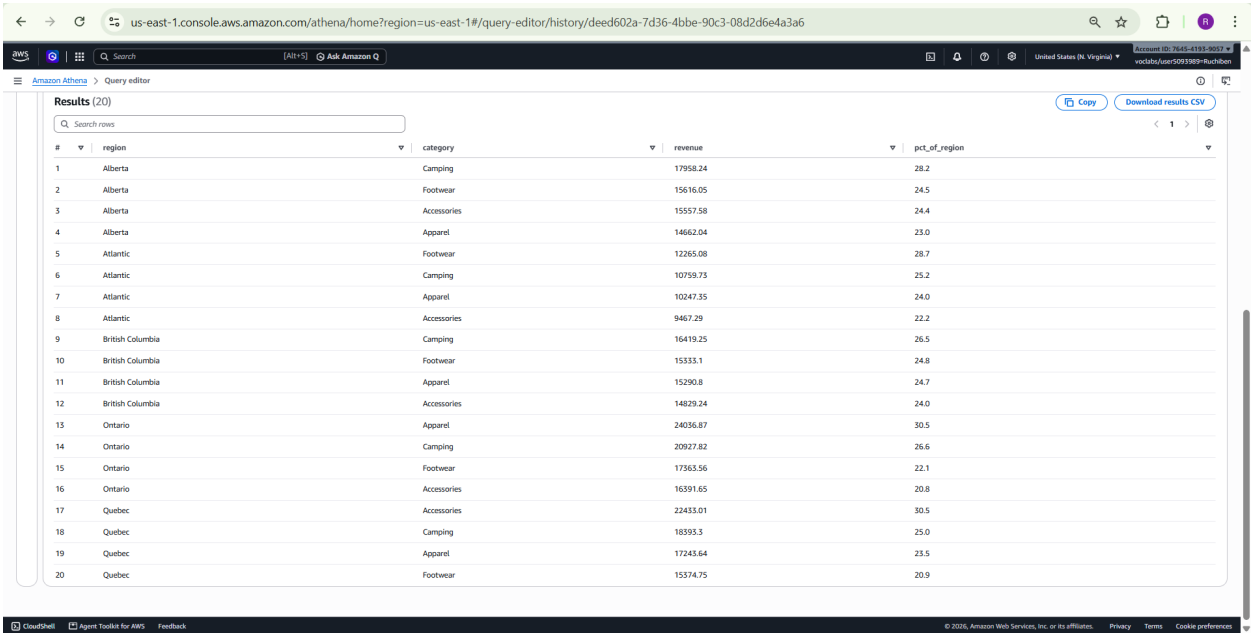


Figure 31. Q2, revenue by region and category

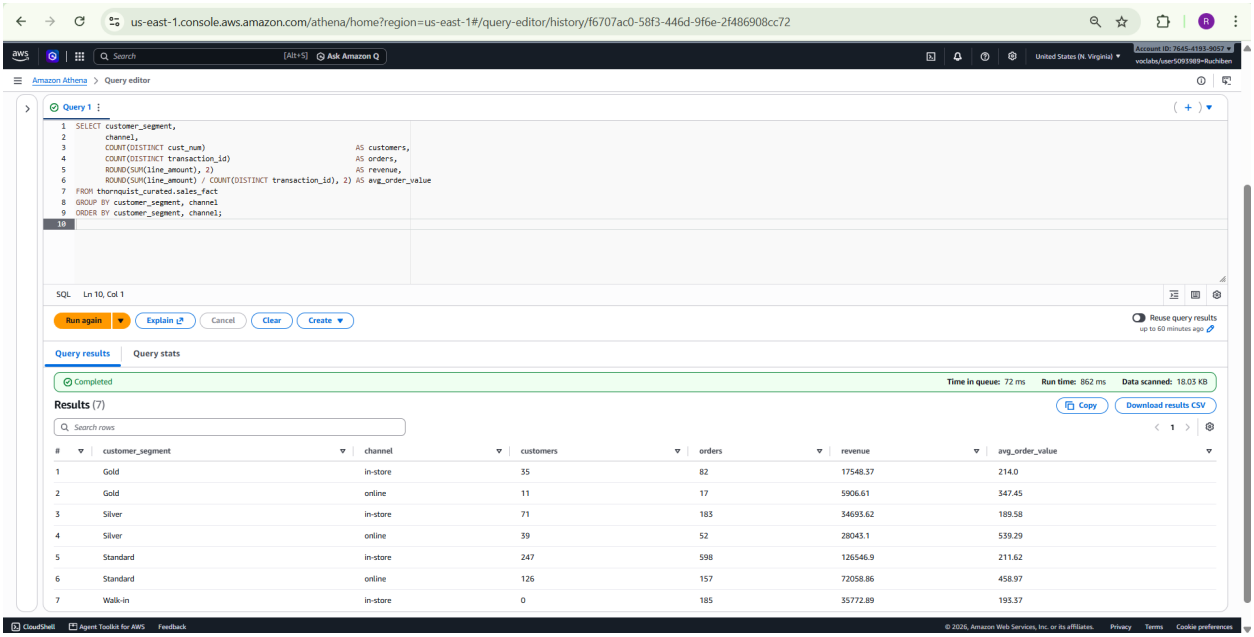


Figure 32. Q3, loyalty segment by channel

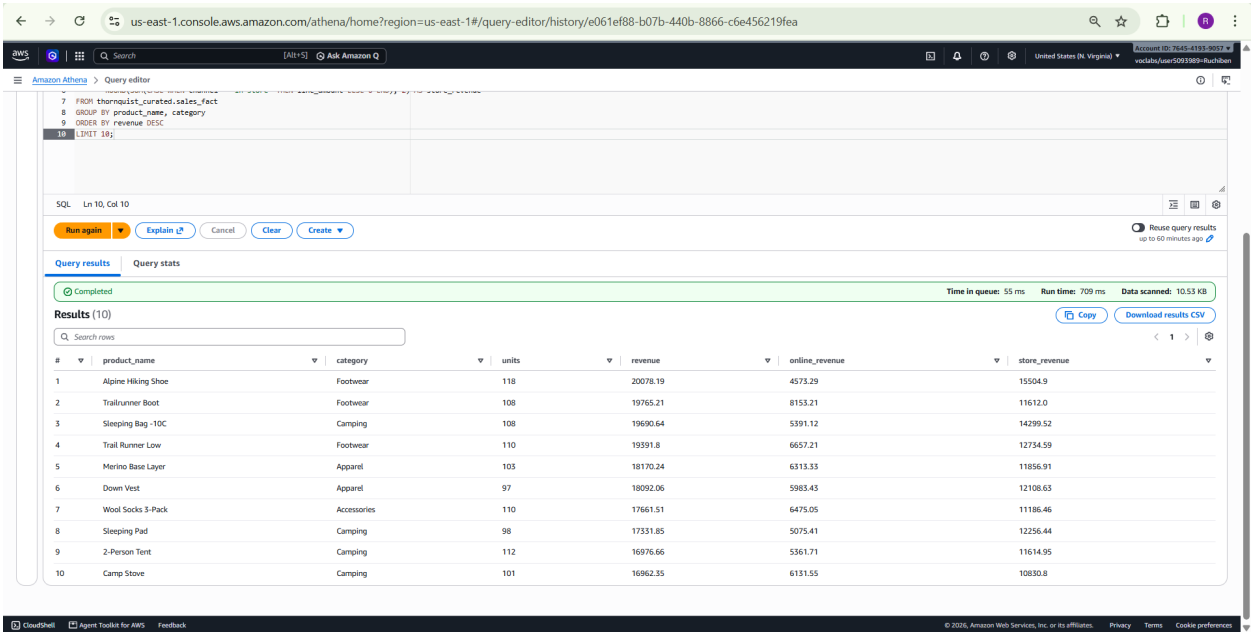


Figure 33. Q4, top products (Alpine Hiking Shoe #1)

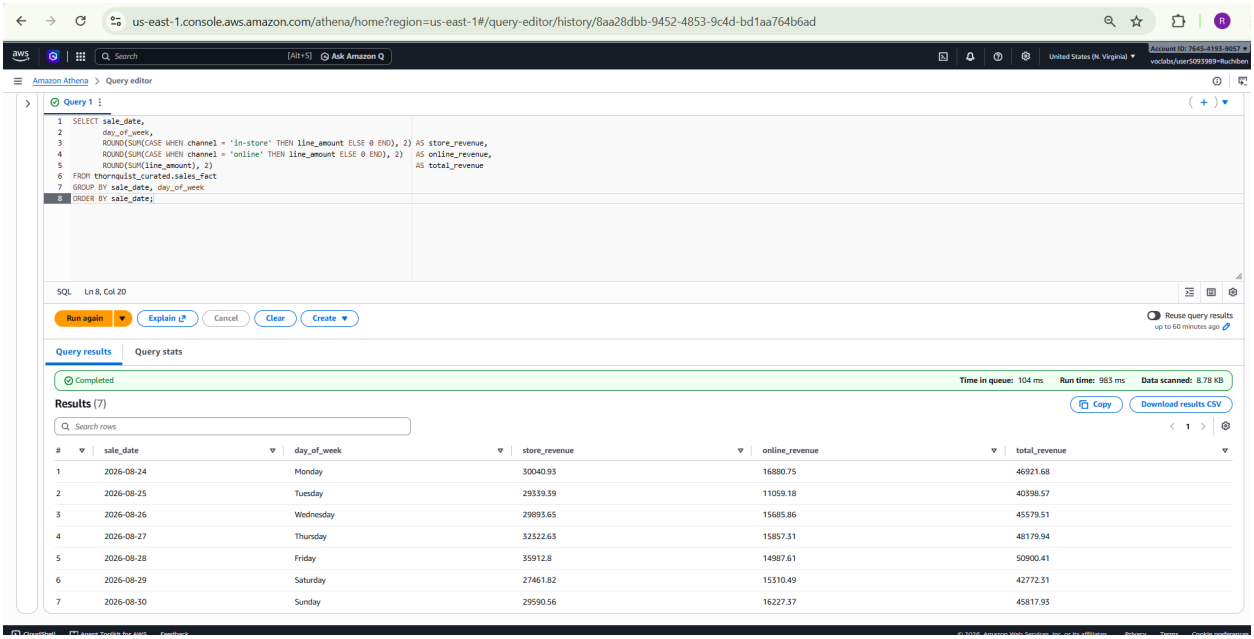


Figure 34. Q5, revenue by day of week

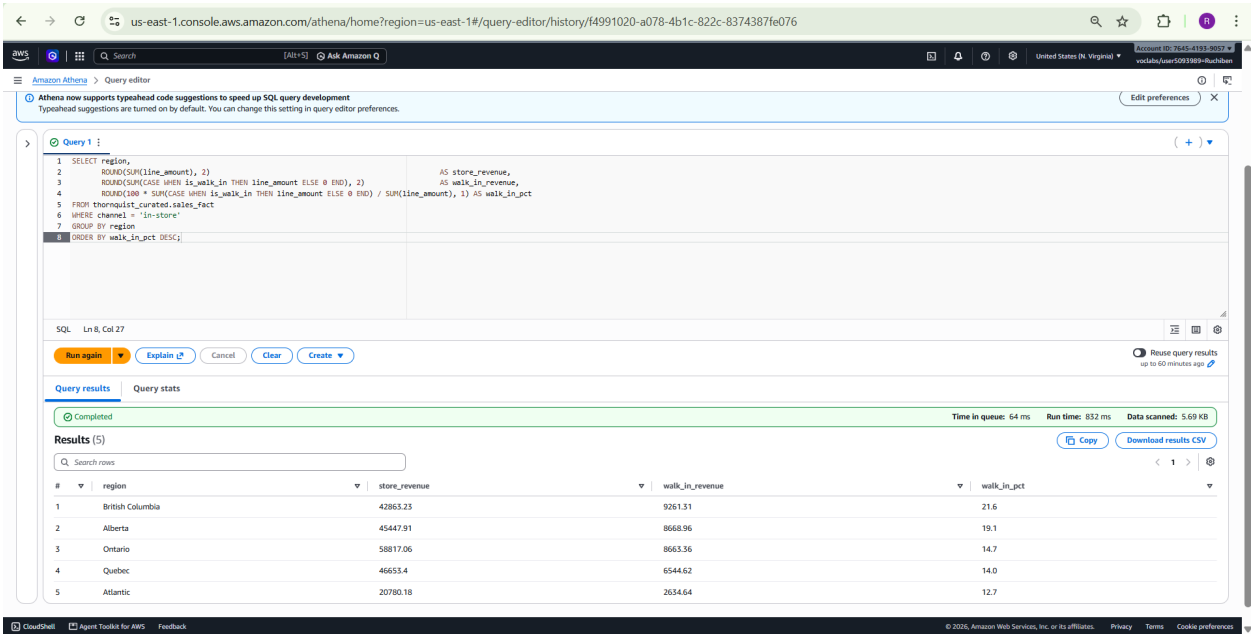


Figure 35. Q6, walk-in share by region (British Columbia highest, 21.6%)

10. Benchmark: CSV vs Parquet (Test 1, online revenue by region)

Same query and same answer on three storage formats; only data scanned changes.

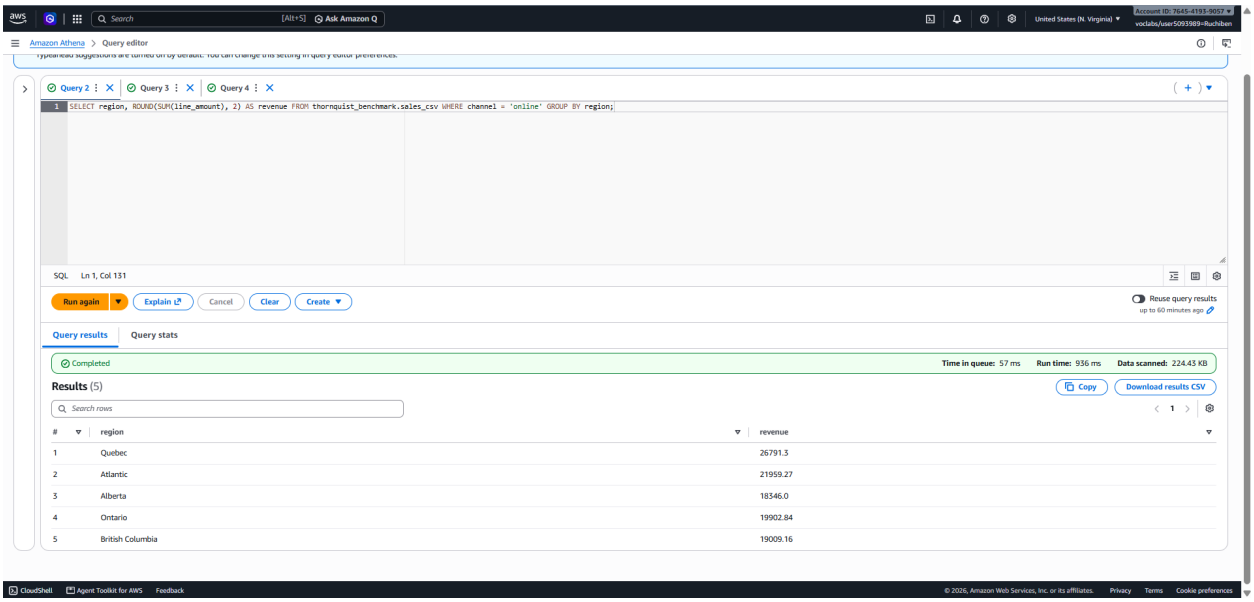


Figure 36. CSV table: 224.43 KB scanned, 936 ms

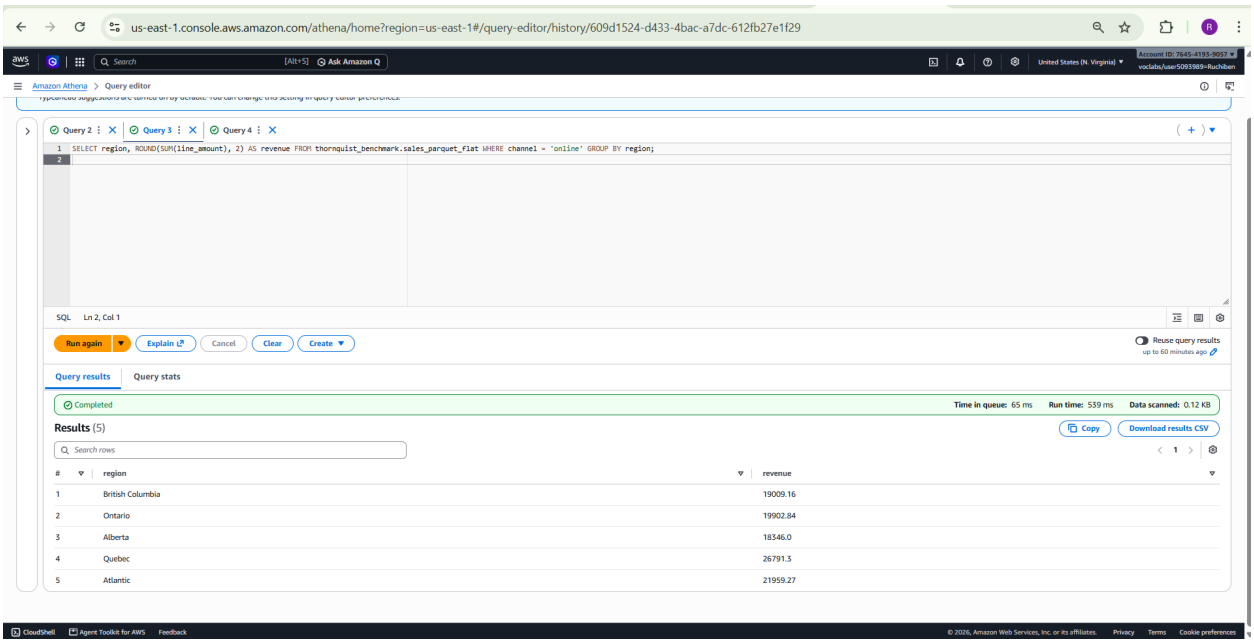


Figure 37. Flat Parquet table: 0.12 KB scanned, 539 ms

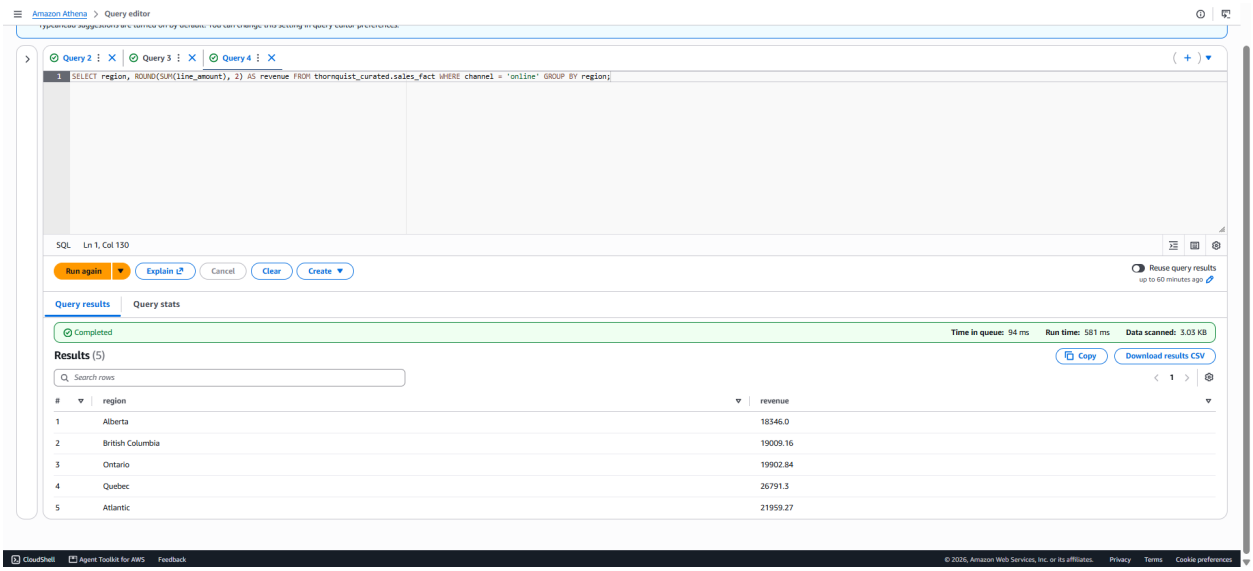


Figure 38. Curated partitioned Parquet: 3.03 KB scanned, 581 ms (98.6% less than CSV)

11. Governance

Controls enforced in AWS.

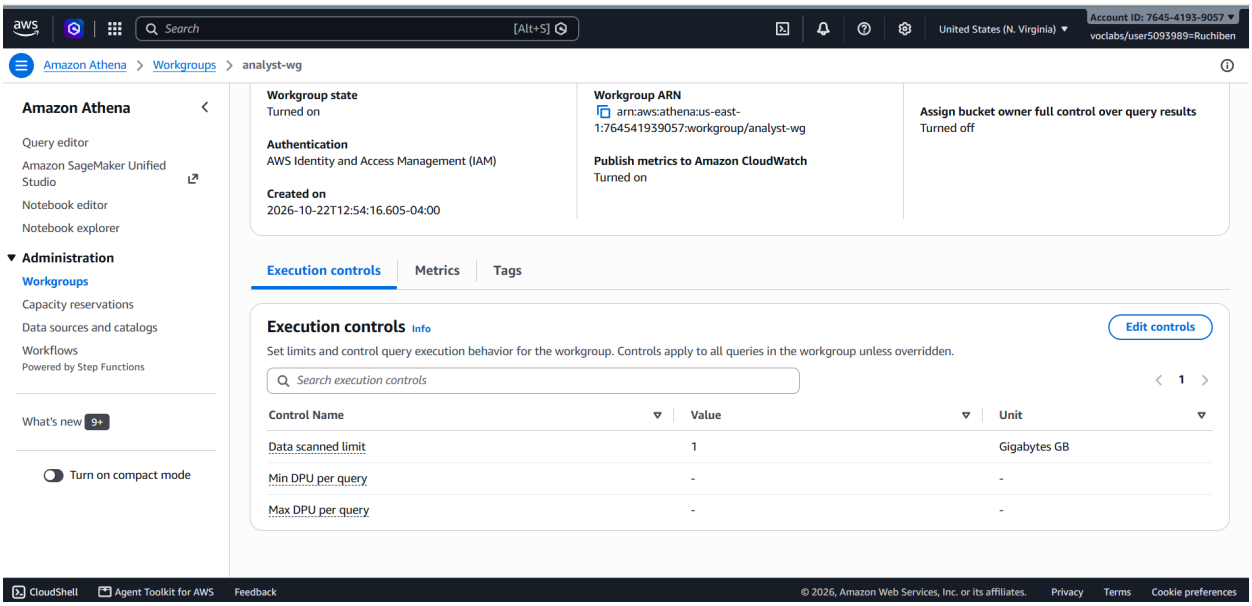


Figure 39. Athena workgroup analyst-wg: 1 GB data-scanned limit per query

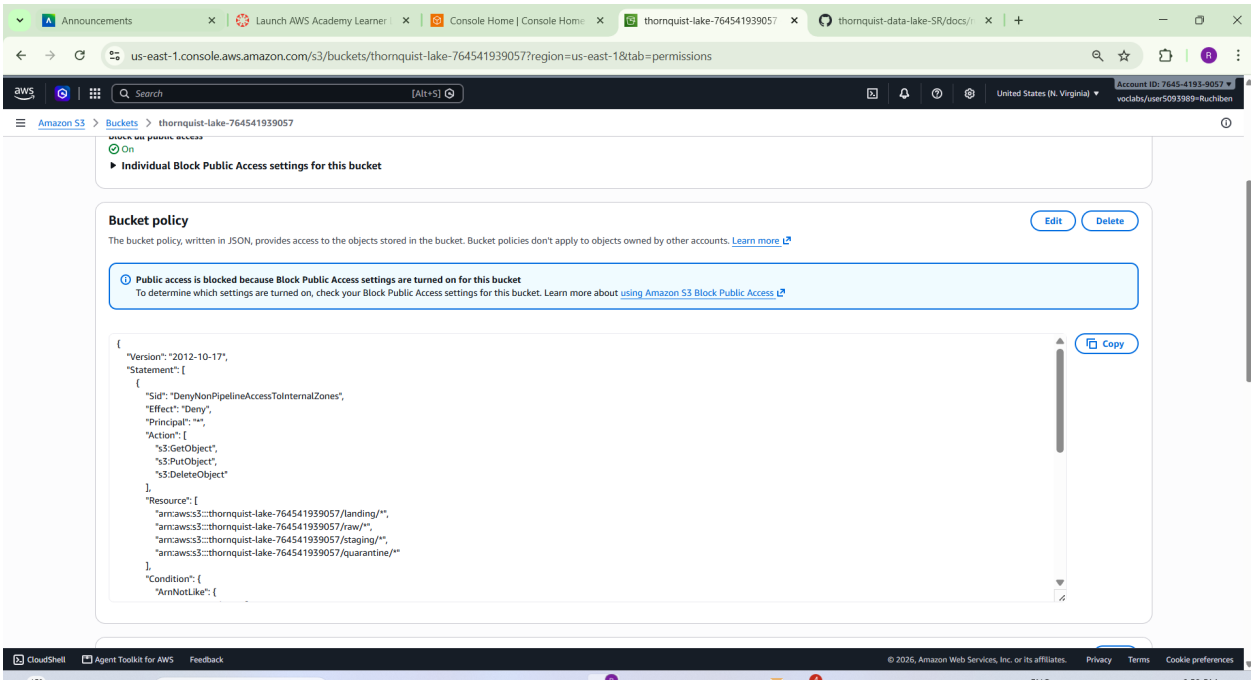


Figure 40. S3 bucket policy: deny access to landing/raw/staging/quarantine except the pipeline role; public access blocked

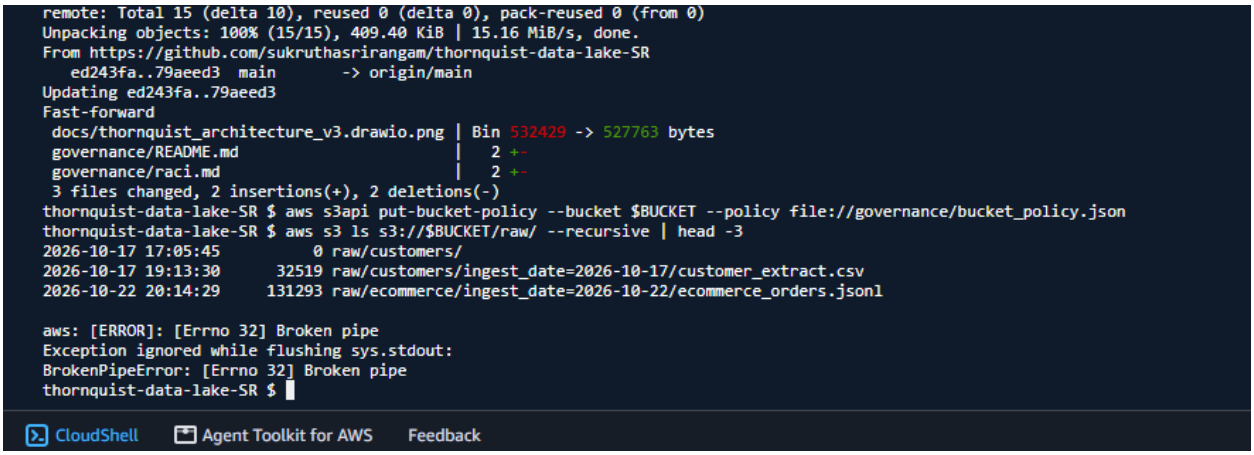


Figure 41. Bucket policy applied from CloudShell; pipeline role can still list raw/

12. Source code

All code pushed to the team GitHub repository.

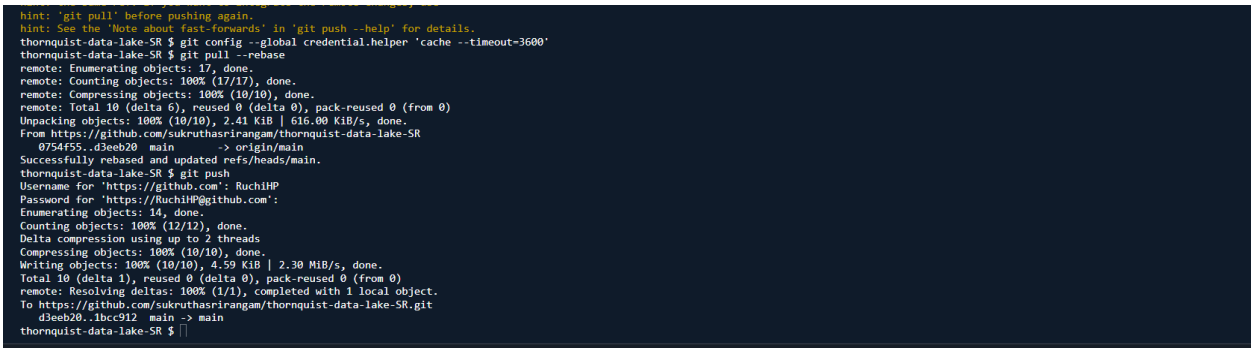


Figure 42. Git push of ingestion, Glue and SQL code to thornquist-data-lake-SR